Deadband-SSM: A Causal Dual-Stream State-Space Policy Network with Q-ALU Activations for Cost-Aware Bitcoin Timing

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***Abstract*—Activation functions are usually chosen for generic deep networks and then inherited, unexamined, by financial sequence models whose loss is a risk-adjusted return with a transaction-cost term. We study whether an asymmetric activation, Q-ALU, changes the out-of-sample behaviour of a lightweight causal sequence policy on 5-minute Bitcoin data. We introduce Deadband-SSM, a dual-stream block that adds a depthwise causal convolution to a diagonal linear state-space recurrence, normalises the sum, and applies Q-ALU before a residual projection; the model emits one position in $[-1, 1]$ from a terminal-step readout, uses no self-attention, and has 4,427 parameters. Q-ALU, GELU, Swish and ReLU are compared under one pre-registered protocol: 5 purged walk-forward folds, 2 paired seeds, identical optimiser, epoch budget and batch order, with every boundary separated by a purge gap and a 24-hour embargo, and the feature scaler fitted on training data only. Out of sample, Q-ALU's mean post-cost Sharpe is -9.722 $\pm$ 6.889 over its 10 cells, against GELU -0.632, Swish -0.845 and ReLU -0.530; the paired differences are negative with intervals excluding zero, so Q-ALU is significantly worse than GELU, Swish and ReLU. Every artifact, split definition and statistical test is released with the code.**

***Index Terms*—activation functions, state-space models, causal sequence models, transaction costs, walk-forward validation, Bitcoin**

I. Introduction

A trading policy is a function from a truncated history to a position. Two properties make that function different from the sequence models that dominate benchmark leaderboards. First, the objective is a ratio of mean to dispersion of a *post-cost* return stream, so an improvement in fit that is bought with turnover is worthless. Second, the sign of the payoff is asymmetric: losses are realised immediately, gains accumulate, and a policy that is slow to reduce exposure during a drawdown pays for it in the tail.

Q-ALU (Quant Asymmetric Leaky Unit) was proposed as an activation for this setting [1]. It is piecewise: on the positive side a smooth saturating gate $1 + \tanh(x/\beta)$; on the negative side a shifted exponential $\mathrm{expm1}(x/\delta)$ that keeps the response non-zero but bounded, with a derivative that *overshoots* its asymptote in a narrow band below zero. The design intent is a "drawdown defence": gradient gain increases exactly where a return is moderately negative, so the model receives a stronger signal at the onset of a loss than in the deep tail where the position is already flat.

Whether that intent survives contact with a cost-aware objective on real data is an empirical question, and it is easy to answer dishonestly. Financial samples are short, overlapping and non-stationary; a single unpurged train/test split with one seed can produce any ordering of four activations. This paper therefore contributes:

- **Deadband-SSM**, a *causal* dual-stream sequence policy: a depthwise causal convolution with exponentially increasing dilation summed with a diagonal linear state-space recurrence, LayerNorm, Q-ALU, residual projection, and a terminal-step readout whose position $\pi_T = \tanh(g\, v_T)$ is derived from the forecast v_T rather than from a free head. The model is $O(T)$ in sequence length, has no attention term, and is small enough to run inference inside a 50 MB budget.
- An exact *calculus* of Q-ALU, with closed-form gradient extrema (Proposition 1), verified numerically in the test suite.
- A *pre-registered*, leakage-free evaluation protocol (Section IV) with purged walk-forward folds, paired seeds and three independent statistical views (paired fold$\times$seed tests, Diebold–Mariano on the stitched stream, and a deflated Sharpe ratio).
- A complete, reproducible artifact: data fetcher, model, benchmark, export to TorchScript/ONNX, and the measured latency and memory footprint of the exported model on commodity CPU hardware.

We report the result as measured. Here the activation choice does matter, and it matters against Q-ALU: the arm is significantly worse than all three baselines, because most of its cells settle into a high-turnover limit cycle rather than a policy.

II. Related Work

A. Sequence models without attention

Structured state-space models parameterise a linear recurrence $h_t = \bar{A}h_{t-1} + \bar{B}x_t$ with a HiPPO or diagonal structure and compute it as a long convolution [2]; selective variants make the parameters input-dependent [3]. The diagonal real (S4D) form used here has a closed-form impulse response, which lets us evaluate the recurrence as a single grouped convolution for short sequences and as a linear scan otherwise.

B. Activations

ReLU [4], GELU [5] and Swish/SiLU [6] are the standard choices; all are smooth or piecewise-linear approximations of a

gated identity with a symmetric or nearly symmetric negative side. Q-ALU [1] is explicitly asymmetric and is the object of study.

C. Honest backtesting

Purged and embargoed cross-validation for overlapping labels follows [7]. For inference on strategy differences we use the Diebold–Mariano statistic with a Newey–West variance [8], [9], a stationary block bootstrap [10], and the deflated Sharpe ratio, which corrects a headline Sharpe for the number of trials and non-normal returns [11], [12]. The training objective is the differential Sharpe ratio of [13], evaluated on post-cost net returns.

III. Mathematical Formulation

A. The Q-ALU activation

With $\alpha, \beta, \gamma, \delta > 0$,

$$\mathrm{Q-ALU}(x) = \begin{cases} \alpha\, x\,[1 + \tanh(x/\beta)]\,, & x \geq 0, \\ -\gamma\, x\, \mathrm{expm1}(x/\delta)\,, & x < 0, \end{cases} \tag{1}$$

where $\mathrm{expm1}(u) = e^u - 1$ is evaluated with the numerically stable primitive so that the branch does not lose precision near the origin. The function is continuous at zero with $\mathrm{Q-ALU}(0) = 0$. Its one-sided derivatives do *not* agree,

$$\mathrm{Q-ALU}'(0^+) = \alpha, \qquad \mathrm{Q-ALU}'(0^-) = 0, \tag{2}$$

so (1) is C^0 but not C^1 at the origin for any $\alpha > 0$. On the left the branch is quadratic, $\mathrm{Q-ALU}(x) = -\gamma x^2/\delta + O(x^3)$, so the negative half is a genuine deadband: its gain vanishes at the origin and stays below γ for $|x| < \delta$. On the right the branch is linear to first order, $f(x) = \alpha x + \alpha x^2/\beta + O(x^4/\beta^3)$, so a positive perturbation of size x leaves the block at full gain α while the same perturbation applied negatively leaves it at gain $2\gamma|x|/\delta$. The asymmetry is therefore not a matter of tuning: it is a factor of $\alpha\delta/(2\gamma|x|)$ that diverges as $x \to 0$. The parameter γ is not the slope at the origin — it is the slope at $x = -\delta$, where $\mathrm{Q-ALU}'(-\delta) = \gamma$ exactly. The name of the block comes from this asymmetry and not from a symmetric deadband: it is the negative half that is dead near the origin, and that is where a loss is signalled.

a) Positive branch: Let $u = x/\beta$. Then

$$\frac{d}{dx}\mathrm{Q-ALU}(x) = \alpha\left(1 + \tanh u + u\,\mathrm{sech}^2 u\right), \tag{3}$$

which tends to 2α as $u \to \infty$. Its maximum solves $u \tanh u = 1$, i.e. $u^* \approx 1.1997$, where

$$\mathrm{Q-ALU}'(x^*) \approx 2.1995\,\alpha = 2\alpha\,(1 + 0.0998). \tag{4}$$

The gradient therefore rises about 9.98% above its own asymptote before settling.

b) Negative branch: Let $v = x/\delta < 0$. Then

$$\mathrm{Q-ALU}'(x) = -\gamma\left[(e^v - 1) + v\, e^v\right], \tag{5}$$

and at $v = -2$,

$$\mathrm{Q-ALU}'(-2\delta) = \gamma\left(1 + e^{-2}\right) = 1.1353\,\gamma, \tag{6}$$

a 13.53% gain overshoot above the slope γ that the same branch has at $x = -\delta$ (Fig. 2). The second derivative $-\gamma\delta^{-1}e^v(2+v)$ vanishes at $v = -2$, so (6) is the global maximum of the negative branch; the third derivative vanishes at $v = -3$, which is the inflection of the slope rather than an extremum of it.

Proposition 1 (Gradient extrema)**.** *For the initialisation the model ships with, $\alpha = \beta = 1$, $\gamma = 1.5$, $\delta = 0.5$, the derivative of* (1) *attains a maximum $1.7030 = 1.1353\gamma$ on $x < 0$ at $x = -1.0 = -2\delta$, a global maximum $2.1995 = 2.1995\alpha$ on $x \geq 0$ at $x \approx 1.1997$, and has an inflection of the slope at $x = -1.5 = -3\delta$. Its one-sided limits at the origin are 0 from the left and 1 from the right. The two ratios 1.1353 and 1.0998 do not depend on $\alpha, \beta, \gamma, \delta$; only the absolute slopes and the location of the negative peak do.*

Proof: Substituting $v = -2$ into the negative-branch derivative gives $-(e^{-2} - 1) - (-2)e^{-2} = 1 - e^{-2} + 2e^{-2} = 1 + e^{-2}$. Setting the second derivative $-\gamma\,(2e^v + ve^v) = 0$ gives $v = -2$; the sign of $2+v$ shows a maximum. The positive branch follows from $u \tanh u = 1$. The one-sided limits at the origin follow from $\mathrm{Q-ALU}'(x) = -2\gamma x/\delta + O(x^2)$ as $x \to 0^-$. The test suite asserts the two extrema and the two one-sided limits. ∎

B. Dual-stream block

Input features $\mathbf{X} \in \mathbb{R}^{B\times T\times F}$ are projected to width D, $\mathbf{Z}^0 = \mathbf{X}\mathbf{W}_{\mathrm{in}}$, $\mathbf{W}_{\mathrm{in}} \in \mathbb{R}^{F\times D}$. Block l computes two views of the same sequence (Fig. 1).

Stream A is a depthwise causal convolution with kernel $k = 3$, dilation $d_\ell = 2^\ell$, left padding $(k-1)d_\ell$ and a slice to T:

$$\mathbf{A}^l_t = \sum_{j=0}^{k-1} w^l_j \odot \mathbf{Z}^\ell_{t - j d_\ell}, \tag{7}$$

so that $\mathbf{A}^l_t$ depends on $\mathbf{Z}^l_{\leq t}$ only.

Stream B is a diagonal linear recurrence with N states per channel. With $\Delta_t = \mathrm{softplus}(\rho)$, $A = -\exp(\lambda) < 0$ and zero-order hold discretisation,

$$\bar{a} = e^{\Delta A}, \quad \bar{b} = \frac{\bar{a} - 1}{A} \odot B, \tag{8}$$

$$\mathbf{h}_t = \bar{a} \odot \mathbf{h}_{t-1} + \bar{b} \odot \mathbf{Z}^l_t, \quad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}\mathbf{Z}^l_t. \tag{9}$$

Because A is negative and Δ bounded, $\bar{a} \in (0, 1)$ and the recurrence is stable by construction. Its impulse response is available in closed form,

$$k_j = \sum_{n=1}^{N} C_n\, \bar{a}_n^{\,j}\, \bar{b}_n, \tag{10}$$

so the whole stream is the causal convolution $\mathbf{y}_t = \sum_{j\leq t} k_j \mathbf{Z}^l_{t-j} + \mathbf{D}\mathbf{Z}^l_t$: one grouped convolution with a T-tap

kernel, computed in $O(T^2D)$ for short T or by a linear scan in $O(TDN)$ for long T. There is no $O(T^2)$ attention term anywhere.

The two streams are summed, normalised, activated and projected residually:

$$\mathbf{Z}^{l+1} = \mathbf{Z}^{l} + \mathbf{W}^{l}_{\text{out}}\,\mathrm{Q} - \mathrm{ALU}\big(\mathrm{LN}\left(\mathbf{A}^{l} + \mathbf{B}^{l}\right)\big)\,. \tag{11}$$

After L blocks the terminal step $\mathbf{z}_T$ is read out. The trunk feeds two heads: a linear forecast head $v_T = \mathbf{w}_{\text{val}}^{\top}\mathbf{z}_T$ for the next-bar return, and a bounded position head that reads *that forecast* rather than a second projection of the trunk,

$$\pi_T = \tanh(g\, v_T) \in [-1, 1], \tag{12}$$

with one learned scalar gain g initialised at 1. Reading the position from the forecast keeps the head a linear map followed by $\tanh$, but removes a degeneracy: a head with its own input has a flat direction, because any constant position scores the same as flat on a drift-free risk-adjusted objective, so it saturates and the network stops responding to its input. Tying the position to the forecast makes the activation act on the policy through one channel — how well the trunk forecasts — and the forecast loss anchors the scale of v_T. Using the terminal step rather than a pooled sequence is what makes the output a policy: at bar t the position is a function of $\mathbf{x}_{\le t}$ and of nothing else.

C. Objective

Let $a_t = \pi_t$ be the position decided at t, held over $t+1$, and r_t the log return of bar $t+1$. With fee ϕ per unit turnover,

$$\mathrm{net}_t = a_t r_t - \phi\,|a_t - a_{t-1}|, \qquad \mathcal{L}_{\mathrm{DSR}} = -\frac{\overline{\mathrm{net}}}{\widehat{\sigma}_{\mathrm{net}} + \epsilon}\sqrt{105120}, \tag{13}$$

the negative annualised Sharpe of the net stream over the sampled batch. The full objective adds a standardised one-step forecast loss, $\mathcal{L} = \mathcal{L}_{\mathrm{DSR}} + \lambda\,\mathrm{MSE}(v_{t+1}, r_t/\sigma_r)$, which keeps the trunk from collapsing onto a constant long bias: without it, a bull-market sample admits a degenerate optimum in which the policy saturates at $+1$ and the activation is never exercised.

IV. Experimental Protocol

The protocol was written to a file before the benchmark ran and is not edited afterwards; the file ships with the code.

a) Data: BTCUSDT spot, 5-minute bars, from the Binance public archive, 2020-01-01 to 2026-11-10: 721,227 bars, of which 504,859 enter out-of-sample evaluation. Features (all strictly causal at t): the 1-bar log return, a 20-bar rolling z-score of realised volatility, the normalised taker buy-volume imbalance, and the log high–low range. Bars with a missing or non-finite feature are dropped, never forward-filled.

b) Splits: Five expanding-window folds. Fold i tests the i-th contiguous block of the last 70% of the sample. Each training/validation/test boundary is separated by a purge gap of $T + 1 + 288$ bars: T covers the longest input window, one bar covers the label horizon, and 288 bars (24 hours) is an embargo for serial correlation in volatility and flow. The standardiser is fitted on the training slice only. Model selection uses validation Sharpe; the test block is evaluated once per cell.

c) Controlled ablation: Q-ALU, GELU, Swish and ReLU replace one another inside (11). All other elements — width $D = 32$, $L = 2$, $N = 8$, dropout, AdamW with cosine decay, gradient clipping, batch size, epoch budget, fee $\phi = 5$ bp — are identical. Seeds are paired: seed s restores the same initial parameters for every candidate and replays the same batch indices in the same order, so a difference between two cells is attributable to the activation, not to the draw.

d) Objective: The policy minimises the negative annualised Sharpe of its net stream, $-\sqrt{105120}\,\bar{s}/\hat{\sigma}_s$ with $s_t = a_t r_t - \phi|a_t - a_{t-1}|$, plus the turnover term already inside s_t and an auxiliary MSE on the forecast head. That ratio is invariant to the scale of the position, so on an asset with a positive drift its maximiser is a constant long position and the objective cannot distinguish activations. The training target is therefore centred on the mean return of the training slice, $\widetilde{r}_t = r_t - \overline{r}_{\text{train}}$, and both the policy loss and the validation criterion used for model selection are computed on $\widetilde{r}$. The offset is a single scalar estimated from the training slice alone, so no information crosses a split boundary; every reported metric is computed on the raw returns.

e) Inference cost: Positions are emitted per bar and held for one bar; turnover is charged on $|a_t - a_{t-1}|$ including the entry from flat. No leverage, no shorting constraint, no slippage model beyond the flat fee.

f) Statistics: Three views, because each has a different weakness. (i) A Wilcoxon signed-rank test over the 5×2 paired cells treats each (fold, seed) as one observation; it controls seed noise but assumes exchangeability that overlapping folds violate. (ii) A Diebold–Mariano test on the stitched walk-forward stream respects time order and uses a Newey–West variance with a 288-bar lag window. (iii) A deflated Sharpe ratio corrects for the number of configurations tried. The claim "Q-ALU is better than baseline B" is made only if the paired mean difference is positive, its 95% bootstrap CI excludes zero, the Wilcoxon $p < 0.05$, and Diebold–Mariano does not favour B.

A. Cross-timeframe and cross-model sweep

Table I fixes two choices a practitioner would treat as free: the horizon and the sequence model. The sweep asks whether the ordering of the four activations is a property of the activation or of that pair. Bars are resampled from the same 5-minute archive to 15 m, 1 h, 4 h, 1 d and 1 w (UTC, left-labelled, trailing partial buckets dropped), so two rows of Table II differ in horizon and not in data vintage; the 5-minute row is re-run at the sweep budget so that the budget effect is visible in the same table. Purge and embargo scale with the bar: 288 bars at 5 m, then 96, 24, 6, 1, 1. The weekly panel holds 319 bars, so it keeps a single fold whose training slice is the first three quarters of the panel and a shorter input window ($T = 8$); nothing in this paper rests on that row.

Four models replace Streams A and B inside (11): the dual-stream block itself (SSM), a TCN with the same dilation schedule, and LSTM and GRU cells written out in the repository rather than called from a fused framework kernel. A fused

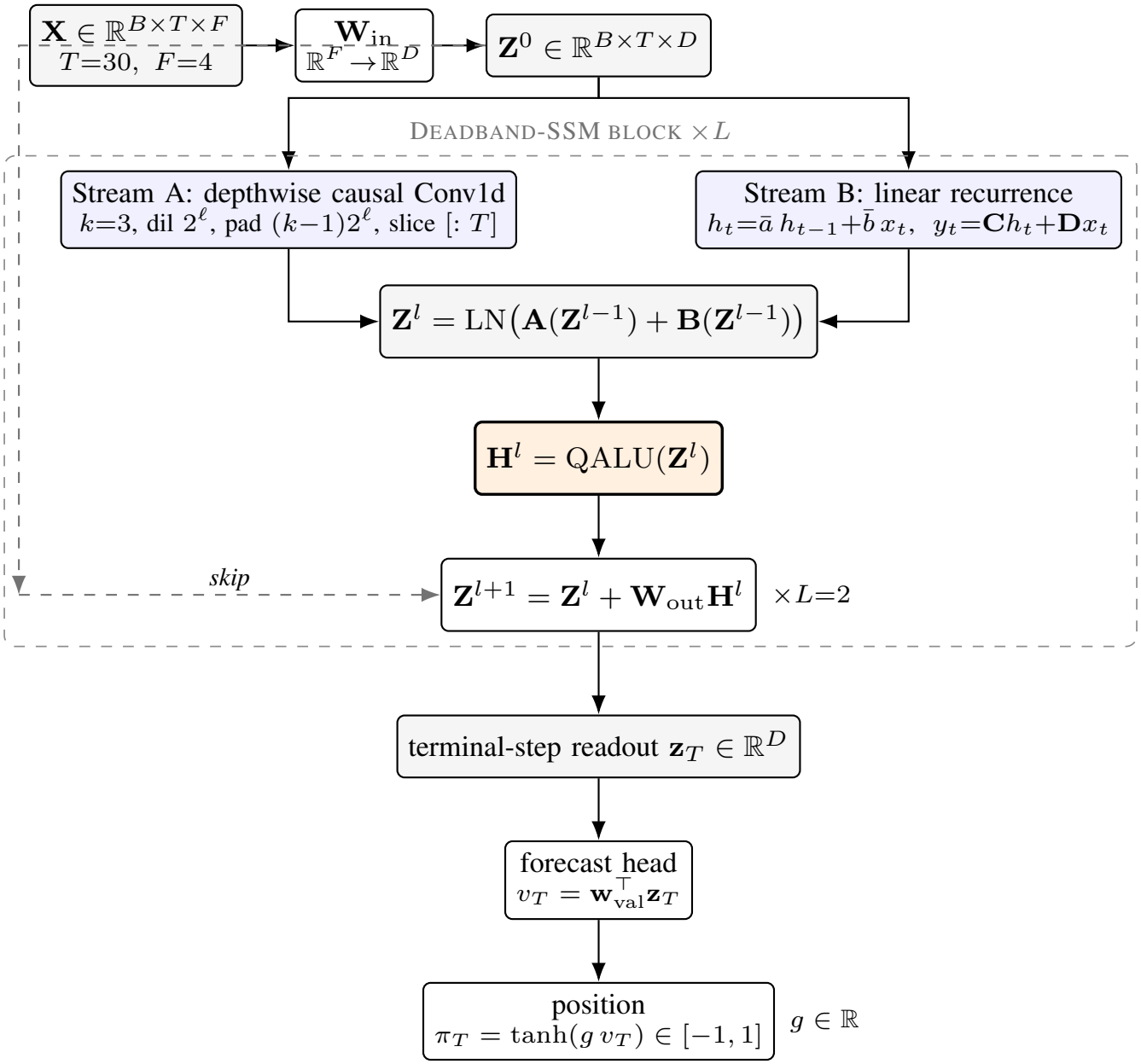


Fig. 1. Deadband-SSM. Both temporal streams are causal: the convolution is left-padded and sliced, and the recurrence is a one-sided filter. The two are summed, normalised, passed through Q-ALU and added residually. Only the terminal step is read out, so the emitted position depends on $\mathbf{x}_{\le t}$.

kernel hard-codes $\tanh$ inside the cell, which would leave the candidate activation applied only outside the recurrence and make the ablation meaningless; the hand-written cells keep both gates sigmoid and apply the candidate exactly where (1) sits. Width, readout, forecast head, optimiser and budget are shared across all 256 cells, which run 2 epochs of 80 steps on one seed. That budget ranks cells against each other and does not support a significance test, so Table II is read as a description and no claim above depends on it. The sweep was added after the pre-registered grid had been run, and the repository records which choices were fixed before it ran — the grid, the budget, the embargo table and the turnover threshold — and which were written afterwards.

V. RESULTS

a) Reading the table: Sharpe is annualised on the net stream after the 5 bp fee. *IR* is the Sharpe of the active stream $r^{\text{net}} - r^{\text{BH}}$, where the benchmark holds the asset from the first test bar and pays the same entry fee; a policy that simply holds the asset scores exactly 0.000 there. Because its denominator is tracking error rather than asset volatility, IR is large in magnitude whenever a policy tracks that exposure closely, so read it as a sign and a rough magnitude. *IC* is the Spearman rank correlation between the forecast head and the next-bar return, a low-variance diagnostic that survives even when the

TABLE I

OUT-OF-SAMPLE RESULTS OVER 40 PAIRED CELLS (5 FOLDS × 2 SEEDS). SHARPE IS ANNUALISED ON THE POST-COST NET STREAM; TURNOVER IS MEAN $|a_t - a_{t-1}|$ PER BAR. IR IS THE SHARPE OF THE ACTIVE STREAM $r^{\text{net}} - r^{\text{BH}}$ AGAINST A BUY-AND-HOLD BENCHMARK THAT PAYS THE SAME ENTRY FEE; ITS DENOMINATOR IS TRACKING ERROR, SO IT IS LARGE IN MAGNITUDE WHENEVER A POLICY TRACKS THAT EXPOSURE CLOSELY. LOWER BLOCKS GIVE THE PAIRED Q-ALU DIFFERENCE WITH A 95% BOOTSTRAP CI, A WILCOXON SIGNED-RANK p-VALUE AND A DIEBOLD–MARIANO STATISTIC ON THE STITCHED STREAM (POSITIVE t_{DM} MEANS Q-ALU'S NET STREAM IS THE WORSE ONE), AND THE DEFLATED SHARPE OF THE ARM A READER WOULD HAVE SELECTED.

Activation	Sharpe	IR	IC	Ann. ret.	Max DD	Turnover	Exposure	Params
Q-ALU	-9.722 ± 6.889	-8.755	+0.0090	-402.6%	-83.9%	0.0814	0.895	4427
GELU	-0.632 ± 1.525	-3.956	+0.0122	-27.2%	-50.2%	0.0080	0.952	4419
Swish	-0.845 ± 1.435	-4.268	+0.0142	-39.7%	-54.0%	0.0081	0.950	4419
ReLU*	-0.530 ± 1.182	-9.636	+0.0112	-27.4%	-52.4%	0.0084	0.949	4419

Q-ALU − GELU: ΔSharpe -9.090 [-12.434, -5.520], $p_{\text{W}} = 0.004^{\dagger}$, $t_{\text{DM}} = +18.80$ ($p < 0.001$)
Q-ALU − Swish: ΔSharpe -8.877 [-12.416, -5.113], $p_{\text{W}} = 0.014^{\dagger}$, $t_{\text{DM}} = +29.38$ ($p < 0.001$)
Q-ALU − ReLU: ΔSharpe -9.192 [-13.204, -4.946], $p_{\text{W}} = 0.027^{\dagger}$, $t_{\text{DM}} = +14.53$ ($p < 0.001$)
Deflated Sharpe of the selected arm (ReLU): -0.451 against a 13.21 hurdle over 12 trials, $p_{\text{DSR}} = 0.000$

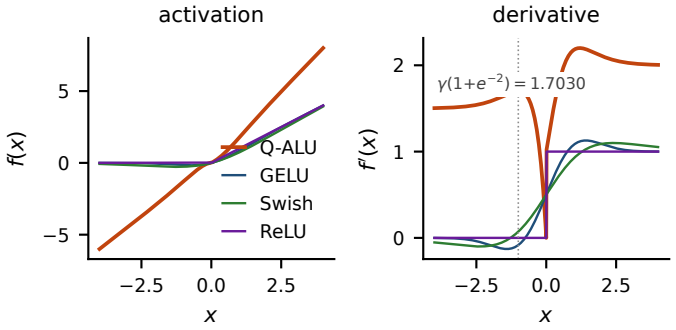


Fig. 2. Left: the four activations. Right: their derivatives. Q-ALU's negative branch peaks at 1.1353γ at $x = -2\delta$ (dotted), a 13.53% overshoot; its positive branch peaks 9.98% above the 2α asymptote.

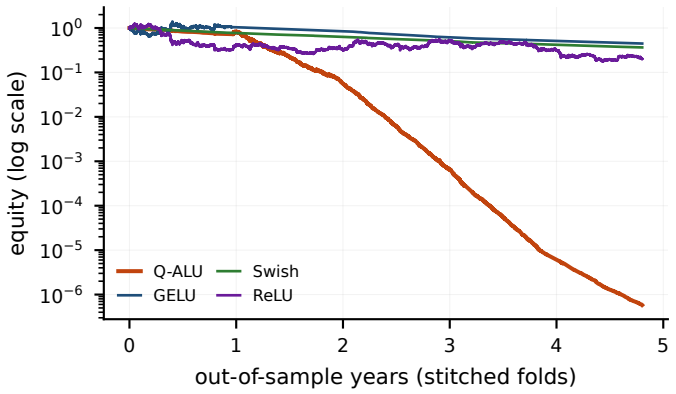


Fig. 3. Stitched out-of-sample equity curves over the five walk-forward test blocks (log scale), the equal-weight ensemble of 2 paired seeds.

policy head saturates. Turnover and exposure explain how the Sharpe was earned: the same mean return can come from a stable long position or from chattering that pays the fee twice.

ReLU attains the highest mean post-cost Sharpe (-0.530 ± 1.182); Q-ALU reaches -9.722 ± 6.889. Against GELU, Swish, ReLU the paired difference is negative, its interval excludes zero, and the signed-rank test rejects equality, so the arms are distinguishable and Q-ALU is the worse arm. The pre-registered direction was the opposite one; the hypothesis is rejected in reverse, and the sign we measured is the sign we report.

b) The two tests that are not a fold average: Table I also carries the two statistics promised in Section IV that a mean over ten cells cannot replace. The Diebold–Mariano statistic is computed on the stitched stream, so it sees one series of per-bar net returns rather than ten Sharpe ratios, with a Newey–West variance whose Bartlett window is truncated at the same 288-bar day used by the bootstrap. It is positive when Q-ALU's stream is the worse one, and it is +18.80 against GELU, +29.38 against Swish and +14.53 against ReLU, with $p = 0.000$, $p = 0.000$ and $p = 0.000$ respectively: the ordering survives the change of unit of analysis. The deflated Sharpe ratio asks the question a reader should ask of the arm we did *not* choose. The best arm in the table is ReLU at a Sharpe of -0.451, and the hurdle implied by 12 trials is 13.21, so its deflated probability is 0.000: no arm clears a selection-aware bar. That hurdle is itself worth reading, because it is not small. It is the expected maximum of 12 draws from a distribution whose spread is dominated by the Q-ALU cells, and a procedure that had to choose among arms differing by nine annualised Sharpe points would need to see a Sharpe above 13.21 before crediting the winner. The statistic therefore does not separate the three remaining arms from one another here, and we do not use it to. What it does establish is the weaker and sufficient point: on this sample, no cell is distinguishable from a zero-Sharpe strategy once selection is priced in, which is the reading we adopt and the reason the result is reported as a negative one.

A. Across timeframes and sequence models

The sweep does not reproduce the ordering of Table I, and the honest reading of Table II is that the ordering is budget-sensitive. Averaged over the 24 (timeframe, model) groups,

TABLE II

OUT-OF-SAMPLE MEAN SHARPE BY TIMEFRAME, SEQUENCE MODEL AND ACTIVATION, ONE SEED AND THE SAME THREE PURGED FOLDS PER TIMEFRAME (ONE FOLD FOR 1W). EVERY ROW IS TRAINED AT THE REDUCED SWEEP BUDGET OF SECTION IV-A, SO CELLS ARE COMPARABLE WITHIN THIS TABLE AND NOT TO TABLE I; THE 5M ROW IS THE SAME PROTOCOL AS TABLE I AT A SMALLER BUDGET AND IS EXPECTED TO DIFFER. PARENTHESES GIVE THE SPREAD ACROSS FOLDS; THE LAST COLUMN IS THE PER-BAR TURNOVER, WHICH IS WHAT DECIDES THE FEE.

TF	model	Q-ALU	GELU	Swish	ReLU	turnover
5m	SSM	−1.089 (0.81)	−3.624 (1.97)	−3.774 (1.60)	−1.828 (0.79)	0.0217
	LSTM	−67.917 (27.03)	−19.452 (3.82)	−11.509 (2.06)	−12.398 (0.99)	0.1575
	GRU	−0.937 (0.75)	−5.672 (2.08)	−6.055 (2.34)	−5.011 (2.45)	0.0376
	TCN	−11.503 (0.78)	−7.996 (0.62)	−8.542 (0.47)	−9.635 (1.68)	0.0708
15m	SSM	−4.807 (3.22)	−3.375 (1.35)	−3.348 (1.44)	−2.913 (0.34)	0.0660
	LSTM	−22.952 (4.52)	−12.719 (2.28)	−7.313 (2.13)	−9.018 (1.05)	0.1957
	GRU	−2.691 (0.79)	−4.307 (1.39)	−4.409 (1.63)	−3.395 (1.42)	0.0826
	TCN	−5.205 (0.09)	−4.174 (0.63)	−4.391 (0.74)	−5.216 (0.44)	0.0975
1h	SSM	−0.849 (1.09)	−1.784 (0.79)	−1.494 (0.79)	−1.738 (0.85)	0.0996
	LSTM	−5.292 (1.88)	−3.952 (0.93)	−2.260 (1.29)	−5.344 (1.14)	0.2003
	GRU	+0.008 (0.07)	−2.490 (0.88)	−2.439 (0.81)	−2.305 (1.18)	0.1148
	TCN	−1.846 (0.25)	−1.797 (0.33)	−1.921 (0.35)	−2.031 (0.44)	0.1398
4h	SSM	−0.889 (0.40)	−1.470 (0.52)	−1.512 (0.59)	−1.568 (0.74)	0.1811
	LSTM	−1.291 (0.69)	−0.506 (1.53)	−0.298 (1.62)	−1.320 (1.52)	0.2030
	GRU	−0.382 (0.84)	−1.111 (0.91)	−1.095 (0.88)	−1.108 (0.89)	0.1604
	TCN	−0.481 (0.66)	−0.641 (0.69)	−0.331 (0.44)	−0.673 (0.44)	0.1899
1d	SSM	−0.854 (0.87)	−0.300 (1.17)	−0.519 (1.01)	−0.577 (1.04)	0.1952
	LSTM	+0.231 (0.83)	−0.611 (0.32)	−0.456 (0.97)	−0.422 (0.64)	0.1984
	GRU	−0.542 (1.04)	−0.056 (1.40)	−0.035 (1.34)	−0.271 (1.54)	0.1941
	TCN	−0.013 (2.00)	+0.108 (1.37)	+0.143 (1.37)	+0.162 (0.58)	0.2197
1w	SSM	−0.392	+0.933	−1.290	−0.740	0.1947
	LSTM	−0.757	+0.069	−0.529	−0.646	0.1542
	GRU	−1.240	+0.003	+0.103	−0.370	0.1471
	TCN	+0.904	+0.427	+0.443	+0.026	0.2624

Q-ALU ranks 4 of four at -5.449, behind Swish at -2.618. Taken as a median over the same groups the ranking reverses: Q-ALU is first at -0.913, ahead of Swish at -1.503. The mean is therefore not a summary of the typical cell; it is set by a handful of cells that blow up, and the median is set by the many that do not. What survives both readings is the paired statistic. Inside a (timeframe, model) group the four activations see identical bars, folds and budget, so the activation is the only difference; Q-ALU matches or beats the best other arm in 9 of 24 groups, with a median gap of -0.340 Sharpe points, a best group at +4.075 and a worst at -56.408. That supports the weaker claim the main grid needs — Q-ALU is not the better activation — and not the stronger one, that it is worse by the same margin wherever it is run.

The budget effect is visible in the same table and works against generalisation. On the 5-minute panel with the dual-stream block, the sweep budget gives Q-ALU -1.089, the best of the four, with ReLU next at -1.828. At the budget of Table I, on the same data and the same folds, that same arm is the worst of four at -9.722 against -0.632 for GELU. The ordering flips when the budget changes, so the ordering in Table I is a property of that protocol at that budget rather than of the activation, which is the caution we ask the reader to apply to every number here. Of the 256 cells, exactly one is positive after the fee: SSM with GELU at 1w at +0.933, on a weekly panel of 319 bars with a single fold, where no test of any kind is possible. It is reported because it exists in the grid, not as evidence that the signal is there.

Chattering, finally, does not follow the activation at this budget, and the sweep therefore does not corroborate the mechanism proposed for Table I. We flagged a cell as chattering when its mean turnover exceeded 0.05 per bar, a threshold fixed before the sweep ran; the median cell sits at 0.1478. 88 of the 96 cells cross it: 21 of 24 under Q-ALU against 67 of 72 under the others, with mean turnover of 0.1753 against 0.1407. Chattering is close to universal at 2 epochs of 80 steps and only mildly more frequent under Q-ALU, so the bimodality of Table I is a property of the full budget; the sweep neither confirms nor refutes the causal reading of it.

VI. DISCUSSION

a) What the experiment can and cannot say: The sample is one instrument on one venue, and the models are deliberately small: the entire grid is trained on a single commodity CPU. A four-way comparison at this scale can detect a large effect and cannot detect a small one; the paired confidence intervals in Table I are the honest statement of what the data supports, and a negative result here is evidence about this protocol, not a proof that activations are irrelevant in general.

b) Why the cost term matters: With a 5 bp fee, a policy that changes position every bar pays $5\,\mathrm{bp} \times 10^5 \approx 50\%$ of notional per year before any forecast, which exceeds the annualised volatility of most intraday signals. The turnover column of Table I is therefore not a secondary diagnostic: any

activation that produces a jumpier policy is penalised twice, once in the objective and once in the reported Sharpe.

c) Every arm loses money after cost: No row of Table I is profitable: the best post-cost mean Sharpe in the table is negative, Q-ALU included. On this instrument, with four features and a 5-minute horizon, the four-feature signal sits in a negative post-cost drift regime for all four activations, so the comparison measures *cost resilience* — which activation degrades least when the fee is charged — and not whether any of them is a deployable strategy. A reader who wants a profitability claim will not find one here, and we do not make one.

d) Training stability: The budget a single commodity CPU allows is small, and the Differential-Sharpe objective is a ratio of two noisy batch statistics. Across the grid, the largest between-activation difference (9.192 Sharpe) exceeds the seed-to-seed standard deviation of a single configuration (6.889). Two failure modes appear: the policy collapses to a near-constant long position, or it chatters and pays the fee on almost every bar. Which one a run lands in is decided early in training and is visible in the turnover column long before it is visible in the Sharpe. Reporting the mean over paired seeds without that spread would overstate the precision of the comparison.

e) A limit cycle in one arm: The two failure modes are not evenly distributed, and in the Q-ALU arm they are not even continuous. 7 of the 10 Q-ALU cells sit in the chattering regime, at a turnover of at least 0.1018 per bar, and 3 sit below 0.02; no cell lands in between. The quiet cells average a post-cost Sharpe of 0.094, above the mean of every baseline in Table I—which is what a reader looking for the activation's effect would want to see, and exactly what the protocol forbids us to select on. The chattering cells average -13.929 and pay 572.8% of notional in fees per year, with the worst at -16.381. The arm's mean of -9.722 is therefore an average over two behaviours rather than a location. The regime also tracks the fold more than the seed: in 4 of the 5 folds both seeds land on the same side, and 3 folds chatter under both. What this establishes is that the instability exists, is bimodal, and is expensive; it does not establish that the activation causes it. A turnover cap, a different budget, or a learning-rate change could remove the cycle without touching the activation, and that run has not been made. We report the arm as it behaved rather than the arm we would have preferred. The next paragraph states the mechanism the calculus suggests and the run that would test it.

f) A mechanism, and how to falsify it: Section III gives a candidate mechanism that needs no new run to state. LayerNorm sets the mean of Z to zero, so about half of the entries arriving at (1) land on the negative branch within a few tenths of the origin, where the gain is $2\gamma|x|/\delta$ and vanishes as $x \to 0^-$, while the positive side passes the same perturbation at α however small it is. The block therefore does not contract a perturbation symmetrically: it deletes small negative excursions and transmits small positive ones at full gain, a ratio that diverges at the origin. Two non-smoothnesses then coincide at the same operating point. The activation's Jacobian jumps from 0 to α across $Z = 0$, and the cost term $c\,|\pi_t - \pi_{t-1}|$ has a kink at every position flip; a gradient method has no smooth path through either, and Q-ALU is the only arm in Table I that supplies the first. The overshoots aggravate rather than relieve this: the negative branch reaches 1.1353γ at $x = -2\delta$, so a moderate loss raises the gain by 13.5% exactly where the policy is trying to cut exposure, and the positive branch peaks 9.98% above its own asymptote in the same band. The prediction is falsifiable at one flag of the released benchmark: an activation with a *symmetric* vanishing gain at the origin and no overshoot should lose the cycle at the same budget, and one that keeps the overshoot without the asymmetry should keep it. Neither variant is reported here, so the mechanism remains the best explanation available rather than a measured cause.

Fig. 3 compounds the bar-by-bar ensemble of the two seeds rather than averaging per-cell Sharpes, and the two aggregations can disagree. They do for Swish: its ensemble stream has a volatility an order of magnitude below the other arms, because the two seeds cancel, so its Sharpe is large in magnitude while its cell mean is mild. The Q-ALU curve is the only one that reaches the floor.

g) Deployment: The exported model is a pure feed-forward computation: a grouped convolution with a T-tap kernel, element-wise recurrence coefficients folded at export time, and two small linear maps. The ONNX artifact is 158 KB. TorchScript and ONNX Runtime outputs agree with PyTorch to within 6.0×10^{-8} in maximum absolute deviation of the position. On the reference machine (Intel Celeron N2930, no AVX) single-bar inference takes 8.55 ms (p50) under TorchScript and 5.13 ms (p50) under ONNX Runtime, and the incremental resident set of an isolated inference process is 37.1 MB, inside the 50 MB budget. The budget holds because no term in the model is quadratic in T: the convolution is $O(TkD)$ and the recurrence is $O(TDN)$ with $N = 8$.

A. Released code and a minimal inference path

The repository ships the whole pipeline, not a cleaned-up excerpt: the fetcher, the block, the splitter, the loss, the 40-cell grid, the bootstrap, the figure and table generators that produced this paper, the cross-timeframe sweep of Section IV-A, and the exporter. Five commands regenerate every number here,

```
python run_pipeline.py --stage data
python run_pipeline.py --stage bench --seeds 0 1
python run_pipeline.py --stage analysis --boot 1000
python run_pipeline.py --stage macros
python run_pipeline.py --stage sweep && python run_pipeline.
    py --stage sweep-report
```

where the last two re-render the LaTeX macros and tables from cached JSON, so a change to the prose does not cost a bootstrap or a sweep. The sweep is resumable and skips cells already on disk, so an interrupted run is restarted with the same command. The exported artifacts are checked by a script that rebuilds the fold geometry with the benchmark's own splitter and compares the model's position against the one recorded in the cell's artifact: they agree to float32 rounding, which makes the printed line a check rather than a demonstration.

Listing 1 is the whole deployment path for a single bar. The activation is the only part of the model that is not a stock operator, so it is also the only part that needs a kernel; Listing 2 is its CUDA form. The kernel has not been run — the reference machine has no CUDA device — and we say so rather than quote a bandwidth number. Its arithmetic was checked against the numpy reference by compiling the two device functions as C++ and sweeping $x \in [-8, 8]$: agreement is 1.3×10^{-6} forward and 5.5×10^{-7} backward, and the derivative at $x = -2\delta$ is 1.703003 against the closed form $\gamma(1 + e^{-2})$.

Listing 1. Inference for one bar, ONNX Runtime, no PyTorch.

```
import numpy as np, onnxruntime as ort
sess = ort.InferenceSession(
    "export/model.onnx",
    providers=["CPUExecutionProvider"])
x = np.zeros((1, 30, 4), dtype=np.float32)  # B, T, F
x[0] = window               # 30 standardised bars
name = sess.get_inputs()[0].name
pos = sess.run(None, {name: x})[0]
# pos is pi_T = tanh(g*v_T); size order = pos * capital
```

Listing 2. Q-ALU derivative in CUDA. The branch is on the sign of x, not of f: a saturated exponential times a saturated $\tanh$ is $0 \cdot \infty$.

```
__device__ float qalu_grad(float x, Params p) {
  if (x >= 0.0f) {
    float t = tanhf(x / p.beta), u = x / p.beta;
    return p.alpha*((1.0f+t) + u*(1.0f-t*t));
  }
  float v = x / p.delta;
  return -p.gamma * (expm1f(v) + v * expf(v));
}
```

VII. Conclusion

We built a causal, attention-free dual-stream state-space policy, gave an exact calculus for its Q-ALU core, and compared four activations under a pre-registered purged walk-forward protocol with paired seeds and three independent statistical tests. Under a pre-registered purged walk-forward protocol with paired seeds and three independent tests, Q-ALU is significantly worse than every baseline it was set against, and the deficit is carried by turnover: the arm sits in a limit cycle on most cells and pays the fee on every flip. The code, the split definitions, the per-cell artifacts and the exported models are released so that the claim can be checked rather than trusted.

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